MA388 Sabermetrics: Lesson 32

Runs Created School

Runs Created Formulas

We encounter three main formulas in *Understanding Sabermetrics* that attempt to quantify a player's offensive production in a season. Each formula gets successively more complicated and is the result of the trend of having more and more offensive statistics available throughout baseball history.

Basic Runs Created

$$RC = \frac{(H + BB) \times TB}{AB + BB}$$

Stolen Base Version of Runs Created

$$RC_{SB} = \frac{(H + BB - CS) \times (TB + (0.55 \times SB))}{AB + BB}$$

"Technical" Version of Runs Created

$$RC_{TECH} = \frac{(H + BB - CS + HBP - GIDP) \times (TB + (0.26 \times (BB - IBB + HBP)) + 0.52 \times (SH + SF +}{AB + BB + HBP + SH + SF}$$

Foundation

All these formulas take on the general formula:

$$RC_X = \frac{A \times B}{C}$$

What do A, B, and C communicate to the equation?

Do we have access to all of these statistics? Yes and no – it depends on the year and whether they were recorded at that time in history yet or not. We do see that Lahman has placeholders for all of them when they are available.

```
library(tidyverse)
library(Lahman)
library(knitr)

Batting |> names()
```

```
 [1] "playerID" "yearID"   "stint"    "teamID"   "lgID"     "G"
 [7] "AB"       "R"        "H"        "X2B"      "X3B"      "HR"
[13] "RBI"      "SB"       "CS"       "BB"       "SO"       "IBB"
[19] "HBP"      "SH"       "SF"       "GIDP"
```

Our goal today is to once again rank offensive production to answer timeless questions like, "Who is the best better in history?" "Which team had the greatest lineup in history?" To get started, we'll compute the *runs created* estimates for each player for each season using all three formulas.

```
RC_df <- Batting |>
  filter(yearID > 1910) |>
  mutate(TB = H + X2B + 2 * X3B + 3 * HR,
         # Basic Runs Created
         RC = (H + BB) * TB / (AB + BB),
         # Runs Created with Stolen Bases
         RC_SB = (H + BB - CS) * (TB + (0.55 * SB)) / (AB + BB),
         # Technical Runs Created
         RC_TECH = (H + BB - CS + HBP - GIDP) *
           (TB + (0.26 * (BB - IBB + HBP)) + 0.52 * (SH + SF + SB)) /
           (AB + BB + HBP + SH + SF))
```

Let's see how many player-seasons yield valid *runs created* estimates using each formula:

```
RC_df |>
  summarize(RC_valid = sum(!is.na(RC)),
            RC_SB_valid = sum(!is.na(RC_SB)),
            RC_TECH_valid = sum(!is.na(RC_TECH))) |>
  kable()
```

RC_valid	RC_SB_valid	RC_TECH_valid
82214	70693	58579

As expected, we have better coverage for the older formulas since some statistics required for the newer formulas have not always been recorded. How good are these formulas? What should the sum of all players' *runs created* be? Let's look at the 2001 season.

How many runs were scored in the American League? The National League?

```
Batting |>
  filter(yearID == 2001) |>
  group_by(lgID) |>
  summarize(totalRuns = sum(R)) |>
  kable()
```

lgID	totalRuns
AL	11013
NL	12186

How many runs were created by all the players in the American League? The National League? Are the *runs created* estimates close to the actual totals?

```
RC_df |>
  filter(yearID == 2001) |>
  group_by(lgID) |>
  summarize(totalRC = sum(RC, na.rm = TRUE),
            totalRC_SB = sum(RC_SB, na.rm = TRUE),
            totalRC_TECH = sum(RC_TECH, na.rm = TRUE)) |>
  kable()
```

lgID	totalRC	totalRC_SB	totalRC_TECH
AL	11211.35	11245.04	11461.6
NL	12748.13	12721.74	12999.9

Not bad. Let's answer our first question.

Question 1: Who is the best batter in history?

You'll need to decide a few things:

- Which formula do you want to use, and which seasons/players does that allow you to consider?
- Are there any eligibility criteria for a player's season to qualify?
- Do you want to look at single-season performance, career performance, or something else?

```
# Make the case for your "best batter" claim.
```

Before we go any further, we need to address an obvious issue with *runs created* estimates: production necessarily requires opportunity, and opportunity is not distributed equally. As we've learned in the course, a batter really has two objectives:

- Do things that lead to runs.
- Don't do things that lead to outs.

We've looked at runs created, but we need to balance that with how many outs a batter has created. The formula for outs created is as follows:

$$\text{Outs} = AB - H + SH + SF + GIDP$$

We see in *Understanding Sabermetrics* a way to balance how many runs a player creates against how many outs they cause. If we imagine a nine-man lineup with just one player, that player has the opportunity to produce runs until reaching 27 outs (three per inning in a nine-inning game). Now consider the average number of runs a player produces over some number of games they've played (total games equals total outs caused divided by 27).

```
RC_df <- RC_df |>
  # Let's consider only players who had some minimum number of at-bats.
  filter(AB > 100) |>
  replace_na(list(SH=0,SF=0,GIDP=0)) |>
  mutate(OUTS = AB - H + SH + SF + GIDP,
         SINGLE_GAMES = OUTS / 27,
         RC_rate = RC / SINGLE_GAMES,
         RC_SB_rate = RC_SB / SINGLE_GAMES,
         RC_TECH_rate = RC_TECH / SINGLE_GAMES,
         .by=c('playerID','yearID'))
```

Question 1 (again): Who is the best batter in history?

Consider now placing a lineup of just one batter against an average opponent. We know the average runs per game created by each player, and we'll compare this number to the league average runs created per game.

```
year_lg_runs_per_game <- Teams |>
  filter(yearID > 1910) |>
  summarize(lg_avg_runs_per_game = sum(R) / sum(G),
            .by=c('lgID','yearID'))

RC_df <- RC_df |>
  left_join(year_lg_runs_per_game, by = c("yearID", "lgID"))
```

Now we can use the Pythagorean Formula to estimate a win percentage for each player.

```
RC_df <- RC_df |>
  mutate(RC_win_pct = RC_rate^2 / (RC_rate^2 + lg_avg_runs_per_game^2),
         RC_SB_win_pct = RC_SB_rate^2 / (RC_SB_rate^2 + lg_avg_runs_per_game^2),
         RC_TECH_win_pct = RC_TECH_rate^2 / (RC_TECH_rate^2 + lg_avg_runs_per_game^2))
```

Again, you'll need to decide a few things:

- Which formula do you want to use, and which seasons/players does that allow you to consider?
- Are there any eligibility criteria for a player's season to qualify?
- Do you want to look at single-season performance, career performance, or something else?

```
# Make the case for your "best batter" claim.
RC_df |>
  filter(AB > 250) |>
  arrange(-RC_rate) |>
  head(5) |>
  select(playerID, yearID, RC,RC_SB,RC_TECH,RC_rate) |>
  kable()
```

playerID	yearID	RC	RC_SB	RC_TECH	RC_rate
bondsba01	2004	183.8033	185.2988	203.3741	20.17353
bondsba01	2002	185.9135	187.6834	208.0899	19.30640

playerID	yearID	RC	RC_SB	RC_TECH	RC_rate
ruthba01	1920	205.8254	200.7835	NA	19.16305
ruthba01	1921	232.8365	228.7498	NA	18.48996
ruthba01	1923	216.2211	208.8958	NA	18.24366

Question 2: Which team had the greatest lineup?

Which team had the best combination of nine batters on its roster in the same year?

```
# Make the case for your "best lineup" claim.
RC_df |>
  group_by(teamID, yearID) |>
  arrange(-RC_win_pct) |>
  slice_head(n=9) |>
  mutate(AVG = mean(RC_win_pct)) |>
  ungroup() |>
  arrange(-AVG) |>
  head(9) |>
  left_join(select(People, playerID, nameFirst, nameLast), by = "playerID") |>
  select(nameFirst, nameLast, yearID, teamID, lgID,RC_win_pct,AVG) |>
  kable()
```

nameFirst	nameLast	yearID	teamID	lgID	RC_win_pct	AVG
Troy	Tulowitzki	2014	COL	NL	0.8688559	0.7347453
Michael	Cuddyer	2014	COL	NL	0.8198842	0.7347453
Corey	Dickerson	2014	COL	NL	0.7986623	0.7347453
Michael	McKenry	2014	COL	NL	0.7833863	0.7347453
Justin	Morneau	2014	COL	NL	0.7485346	0.7347453
Drew	Stubbs	2014	COL	NL	0.6988568	0.7347453
Nolan	Arenado	2014	COL	NL	0.6841895	0.7347453
Charlie	Blackmon	2014	COL	NL	0.6368969	0.7347453
Wilin	Rosario	2014	COL	NL	0.5734411	0.7347453

If you found what I found, here is some historical perspective.